

NEUTRON STARS

A NEUTRON STAR IS AN EXCEEDINGLY DENSE, HOT STAR REMNANT, FORMED FROM THE COLLAPSE OF THE CORE OF A MUCH LARGER STAR—FOUR TO EIGHT TIMES MORE MASSIVE THAN THE SUN—IN A SUPERNOVA EXPLOSION.

Neutron stars are tiny—only about 7–15 miles (10–25 km) across, or about the size of a large city. They are so dense that if a piece the size of a grain of sand was brought to Earth, it would weigh the same as a large passenger airplane. Because they are so compact, neutron stars produce extremely strong gravity: an object on a neutron star's surface would weigh 100 billion times more than on Earth. Whereas normal matter is made of atoms—which contain a lot of empty space—neutron stars consist of much more compact matter, mainly the subatomic particles called neutrons.

Surface

A neutron star's gravity is so strong that its solid surface, which is a million times stronger than steel, is pulled into an almost perfectly smooth sphere

Radiation beam

Neutron stars produce beams of electromagnetic radiation from their magnetic poles

Axis of rotation
Neutron stars spin rapidly, some as fast as 700 times per second

Magnetic field

A neutron star has an extremely powerful magnetic field, which rotates at the same speed as the star

△ Features of a neutron star

A neutron star is an extremely dense, spherical, spinning object, with a surface temperature of about 1,080,000°F (600,000°C). The surface is extremely smooth, its highest "mountains" being no more than $\frac{1}{8}$ in (5mm) tall. Neutron stars produce beams of electromagnetic radiation, which can be light, radio waves, X-rays, or gamma rays.

